Section: 150UB14.0   Span: 12.50 m

PASS

Inputs

Span: 12.50 m

Tributary width: 3.00 m

Restraint: simple, simple FF (Le mult 1.00)

Loads:

Section Properties

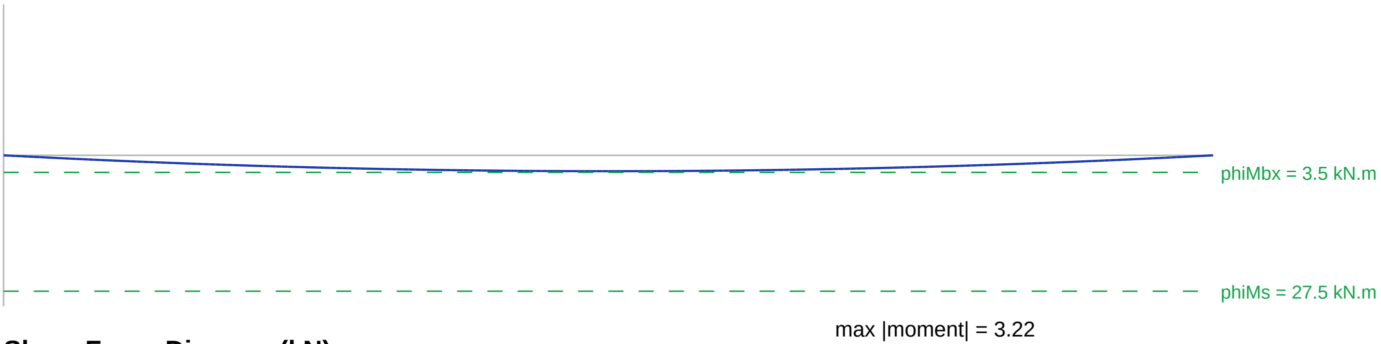
Designation	150UB14.0
Mass	14.0 kg/m
d	150.0 mm
bf	75.0 mm
tf	7.00 mm
tw	5.00 mm
Ix	6.66e+6 mm^4
Sx	1.02e+5 mm^3
Zx	88800.00 mm^3
Iy	4.95e+5 mm^4
J	27000.00 mm^4
Iw	2.53e+9 mm^6
fy	300 MPa

Design Check Summary

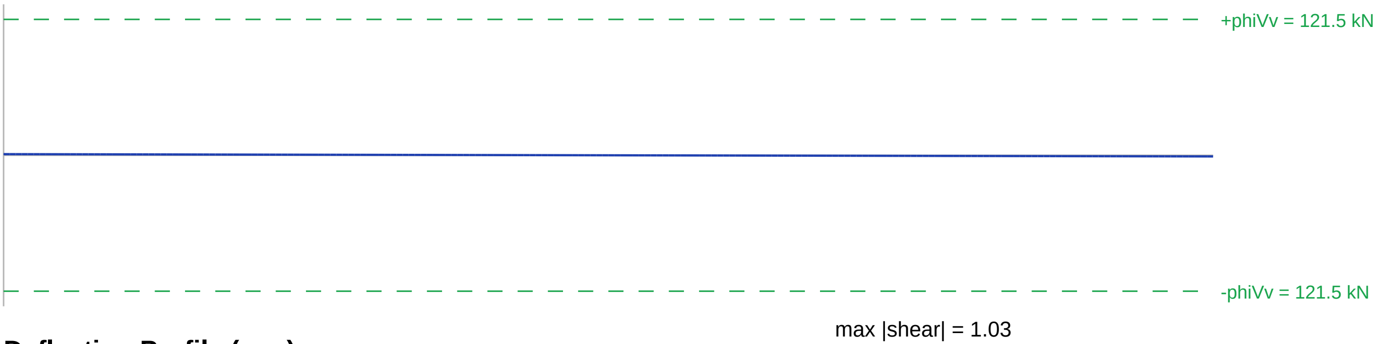
Check	Demand	Capacity	Util	Status
Section moment	3.2 kN.m	27.5 kN.m	11.7%	PASS
Member moment	3.2 kN.m	3.5 kN.m	93.0%	PASS
Shear	1.0 kN	121.5 kN	0.8%	PASS
Deflection G+0.4Q	32.8 mm	41.7 mm	78.7%	PASS
Deflection G	32.8 mm	34.7 mm	94.4%	PASS

Diagrams

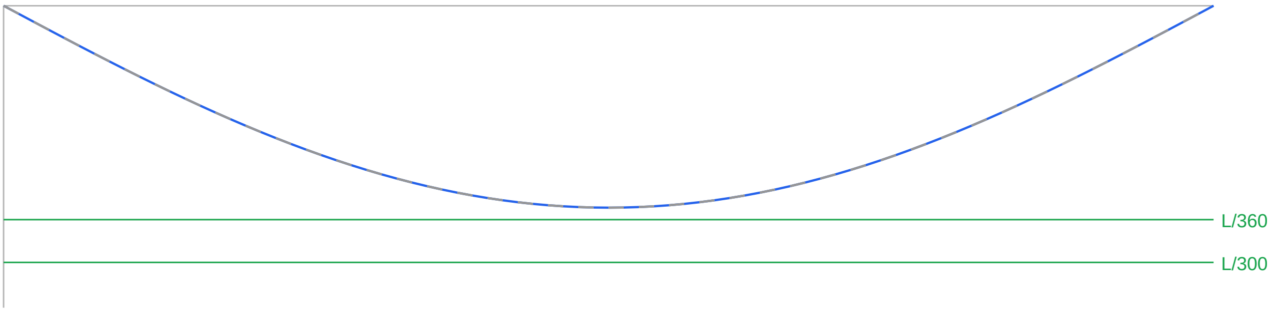
Bending Moment Diagram (kN.m)



Shear Force Diagram (kN)



Deflection Profile (mm)



1. Material Properties [AS4100 Cl. 2.1]

fy = 300 MPa (tf = 7.0 mm)

2. Section Classification - Flange [AS4100 Cl. 5.2.2]

lambda_f = (bf / 2tf) x sqrt(fy/250) = 5.48

Compact limit lambda_ep = 9 -> lambda_f <= lambda_ep -> Flange: Compact

3. Section Classification - Web [AS4100 Cl. 5.2.2]

lambda_w = (d - 2tf)/tw x sqrt(fy/250) = 29.80

Compact limit lambda_ep = 82 -> lambda_w <= lambda_ep -> Web: Compact

Overall section class: Compact

4. Effective Section Modulus [AS4100 Cl. 5.2.3]

Ze = 1.02e+5 mm^3 (per Compact section)

5. Section Moment Capacity [AS4100 Cl. 5.1]

Msx = Ze x fy = 1.02e+5 x 300 = 3.06e+7 N.mm

phiMs = 0.9 x Msx / 1e6 = 27.5 kN.m

6. Effective Length & Supports [AS4100 Cl. 5.6.3]

Support: Pin-Pin

Le = 12.50 m

7. Reference Buckling Moment [AS4100 Cl. 5.6.1.1]

Moa = (pi/Le) x sqrt(E.Iy x (G.J + pi^2.E.Iw/Le^2)) = 3.7 kN.m

8. Member Slenderness Reduction [AS4100 Cl. 5.6.1]

alpha_m = 1.17 (moment distribution factor)

alpha_s = 0.6 x (sqrt((Msx/Moa)^2 + 3) - Msx/Moa) = 0.108

9. Member Moment Capacity [AS4100 Cl. 5.6]

phiMbx = min(0.9 x alpha_m x alpha_s x Msx, phiMs) = 3.5 kN.m

10. Shear Capacity [AS4100 Cl. 5.11.4]

Aw = 750 mm^2

d/tw = 30.0 <= 82.sqrt(250/fy) = 74.9 -> no web slenderness reduction

phiVv = 0.9 x 0.6 x fy x Aw / 1000 = 121.5 kN

11. Load Combinations [AS1170.0 Cl. 4.2.2]

ULS governing: 1.2G+1.5Q -> Mmax = 3.2 kN.m, Vmax = 1.0 kN

SLS combo: G + psi_l x Q (psi_l = 0.4, Office)

12. Deflection Check - G+psi_l.Q [AS1170.1 Cl. 4 / AS4100 Cl. 1.5.3]

Live load type: Office -> psi_l = 0.4

delta = 32.8 mm <= L/300 = 41.7 mm -> PASS

13. Deflection Check - G only [AS4100 Cl. 1.5.3]

delta = 32.8 mm <= L/360 = 34.7 mm -> PASS